

LAST NAME: _____

Practice Quiz 04C — MATH 2055 — Fall 2026

Practice for Quiz 04

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let a, b, c be three distinct objects, and let

$$\mathcal{F} = \{\{a\}, \{a, b\}, \{a, c\}\}.$$

- (a) Compute $\bigcup \mathcal{F}$ and $\bigcap \mathcal{F}$.

- (b) Decide whether $\{a\} \in \mathcal{F}$ and whether $\{\{b\}\} \subseteq \mathcal{F}$. Explain each answer using the elements listed in $\mathcal{F}$.